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The Editors
Experimental Mathematics

Dear Editors,

Please consider the enclosed manuscript, "A SAT-Modulo-Symmetries verification of the Erdős–Gyárfás power-of-two cycle conjecture for minimum-degree-3 graphs up to 30 vertices," for publication in *Experimental Mathematics*.

The Erdős–Gyárfás conjecture (1995) states that every graph of minimum degree at least 3 contains a cycle whose length is a power of two. It is open. The strongest previously published computer search establishes that any *general* minimum-degree-3 counterexample has at least 17 vertices (Royle and Markström), and that any *cubic* counterexample has at least 30 vertices (Markström, 2004), a frontier untouched for two decades.

This manuscript reports a new computational result: using SAT Modulo Symmetries together with the Glasgow subgraph solver as a complete forbidden-subgraph propagator, we verify the conjecture for *all* minimum-degree-3 graphs on at most 30 vertices, raising the general lower bound on a counterexample from 17 to 31. To our knowledge, confirmed by an adversarial literature search, this is the first application of SAT-based methods to this conjecture.

We believe the work fits *Experimental Mathematics* well: it is a computer-assisted result that advances a concrete frontier, with an explicit, reproducible methodology and a multi-layer verification protocol (an exact ground-truth check against `nauty`, reproduction of the established baseline, cross-validation by an independent solver, and robustness across encodings and symmetry-breaking methods). All code and data are openly available, and we discuss the path to a fully machine-checked proof certificate.

The manuscript is original, is not under consideration elsewhere, and has not been previously published. In accordance with the journal's policy, the manuscript includes a disclosure of the generative-AI assistance used in its preparation; the author takes full responsibility for all content and claims.

Thank you for your consideration.

Sincerely,

Arjun Balaji
Columbia University